

MODEL QUESTION PAPER 1

S6 INSTRUMENTATION ENGINEERING

SUBJECT: ENVIRONMENTAL INSTRUMENTATION

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	List any two portable analytical instruments.	M1.02	R
2	Write the principle of gas chromatography. List any two sources of water pollution.	M1.02	R
3	List any two sources of water pollution.	M2.01	R
4	Mention any two water quality parameters.	M2.02	U
5	Write two methods of waste water recycling.	M3.01	U
6	List any two rain water harvesting methods.	M3.03	U
7	List any two measures to control noise pollution.	M4.02	U
8	Name any two gases causing air pollution. S.	M4.01	R
9	Define air pollution.	M4.01	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Explain sensor requirement for environment monitoring.	M1.01	U
2.	Explain Thermal conductivity detectors.	M2.02	R
3.	Write short notes on water treatment objectives and considerations	M2.03	U
4.	Describe the laboratory analysis of ground water samples.	M2.04	U
5.	Explain Chemical Oxygen Demand (COD).	M3.01	U
6.	Describe open channel waste water flow measurement.	M3.02	U
7.	Explain necessity of Rain water harvesting.	M3.03	R
8.	Write short notes on control of air pollution.	M4.01	R
9.	Explain the working principle of Ceilometer in weather stations.	M4.03	U
10.	Write short notes on REMS.	M4.03	U

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Describe the necessity of Instrumentation & control for environment monitoring	M1.01	U
	OR		
IV	Explain the working of Ultraviolet analyzers used for environment monitoring.	M1.02	U
V	Describe pH analysers.	M2.02	U
	OR		
VI	Write short notes on ground water level measurements	M2.04	U
VII	Describe opacity monitors	M2.02	U
	OR		
VIII	Write short notes on drinking water quality standards	M2.01	R
IX	Describe the Instrumentation set up for waste water treatment plant	M3.01	U
	OR		
X	List the various roles of NGOs & municipal corporation in rain water harvesting	M3.03	U
XI	Write short notes on Automatic waste water sampling	M3.01	U
	OR		
XII	Write short notes on rain water harvesting methods	M3.03	R
XIII	Explain the analytical methods for air pollution studies.	M4.01	U
	OR		
XIV	Describe any two effects of sound pollution in environment.	M4.02	R

Question Wise Analysis

Course: **ENVIRONMENTAL INSTRUMENTATION** – Question -1

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.02	R	1	2
I.2	M1.02	R	1	2
I.3	M2.01	R	1	2
I.4	M2.02	U	1	2
I.5	M3.01	U	1	2
I.6	M3.03	U	1	2
I.7	M4.02	U	1	2
I.8	M4.01	R	1	2
I.9	M4.01	R	1	2
II.1	M1.01	U	3	8
II.2	M2.02	R	3	8
II.3	M2.03	U	3	8
II.4	M2.04	U	3	8
II.5	M3.01	U	3	8
II.6	M3.02	U	3	8
II.7	M3.03	R	3	8
II.8	M4.01	R	3	8
II.9	M4.03	U	3	8
II.10	M4.03	U	3	8
III	M1.01	U	7	16
IV	M1.02	U	7	16
V	M2.02	U	7	16
VI	M2.04	U	7	16
VII	M2.02	U	7	16
VIII	M2.01	R	7	16
IX	M3.01	U	7	16
X	M3.03	U	7	16
XI	M3.01	U	7	16

XII	M3.03	R	7	16
XIII	M4.01	U	7	16
XIV	M4.02	R	7	16

Course: **ENVIRONMENTAL INSTRUMENTATION –QP-1**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	8	16.5 ± 6	2	2	1	3	2	14	5	19
2	20	41 ± 6	2	2	3	9	4	28	9	39
3	20	41 ± 6	2	2	3	9	4	28	9	39
4	12	24.5 ± 6	3	3	3	9	2	14	8	26
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	35	28.5
Understanding	88	71.5
Applying	0	0
Analysing	0	0
Evaluating	0	0

Creating	0	0
Total	123	100

MODEL QUESTION PAPER 2

S6 INSTRUMENTATION ENGINEERING

SUBJECT: ENVIRONMENTAL INSTRUMENTATION

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	Mention any two advantage of portable analytical instruments.	M1.02	U
2	List any two stationary analytical instruments.	M1.02	R
3	Mention any two water sources.	M2.01	R
4	List any two instruments used in ground water monitoring.	M2.04	R
5	Mention any two latest methods of waste water treatment plants.	M3.01	U
6	List any waste water measurement techniques.	M3.01	U
7	List any two air sampling methods.	M4.01	R
8	Mention any two effect of sound pollution to environment	M4.02	U
9	List any two instruments used in weather station.	M4.03	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Mention any three needs of instrumentation for environment monitoring.	M1.01	U
2.	Describe the working principle of ultraviolet analyzer.	M1.02	R
3.	Write short notes on opacity monitor.	M2.02	R
4.	Describe the level measurement in ground water monitoring wells.	M2.04	R
5.	Explain BiochemicalOxygenDemand (BOD).	M3.01	U
6.	Sketch any instrumentation setup for waste water treatment plant	M3.01	R
7.	Describe the role of NGO's in rain water harvesting.	M3.03	U
8.	Explain any analytical method for air pollution study.	M4.01	U
9.	Write short notes on importance of air pollution.	M4.01	U
10.	Explain the working principle of Barometer in weather stations	M4.03	R

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Briefly explain Gas chromatography.	M1.02	R
	OR		
IV	Explain the block diagram of portable gas analyzer.	M1.02	R
V	Describe the working of Thermal conductivity detector.	M2.02	R
	OR		
VI	Explain laboratory analysis of ground water samples.	M2.04	U
VII	Explain methods of ground water treatment.	M2.03	U
	OR		
VIII	Briefly explain instrumentation in assessment of soil and ground water pollution	M2.04	U
IX	Describe the waste water measurement techniques.	M3.01	U
	OR		
X	Write short notes on Non open channel flow measurement.	M3.02	U
XI	Explain the methods for rain water harvesting.	M3.03	U
	OR		
XII	Write short notes open channel flow measurement.	M3.02	U
XIII	Write short notes on REMS.	M4.03	R
	OR		
XIV	Describe the methods to control air pollution.	M4.01	U

Question Wise Analysis

Course: **ENVIRONMENTAL INSTRUMENTATION** – Question -2

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.02	U	1	2
I.2	M1.02	R	1	2
I.3	M2.01	R	1	2
I.4	M2.04	R	1	2
I.5	M3.01	U	1	2
I.6	M3.01	U	1	2
I.7	M4.01	R	1	2
I.8	M4.02	U	1	2
I.9	M4.03	R	1	2
II.1	M1.01	U	3	8
II.2	M1.02	R	3	8
II.3	M2.02	R	3	8
II.4	M2.04	R	3	8
II.5	M3.01	U	3	8
II.6	M3.01	R	3	8
II.7	M3.03	U	3	8
II.8	M4.01	U	3	8
II.9	M4.01	U	3	8
II.10	M4.03	R	3	8
III	M1.02	R	7	16
IV	M1.02	R	7	16
V	M2.02	R	7	16
VI	M2.04	U	7	16
VII	M2.03	U	7	16
VIII	M2.04	U	7	16
IX	M3.01	U	7	16
X	M3.02	U	7	16
XI	M3.03	U	7	16
XII	M3.02	U	7	16
XIII	M4.03	R	7	16
XIV	M4.01	U	7	16

Course: **ENVIRONMENTAL INSTRUMENTATION –QP 2**

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Mark Distribution

Module	hr / Module	Marks/Module ($\sum h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	8	16.5 ± 6	2	2	2	6	2	14	5	22
2	20	41 ± 6	2	2	2	6	4	28	9	36
3	20	41 ± 6	2	2	3	9	4	28	9	39
4	12	24.5 ± 6	3	3	3	9	2	14	8	26
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	38	30.9
Understanding	85	69.1
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100